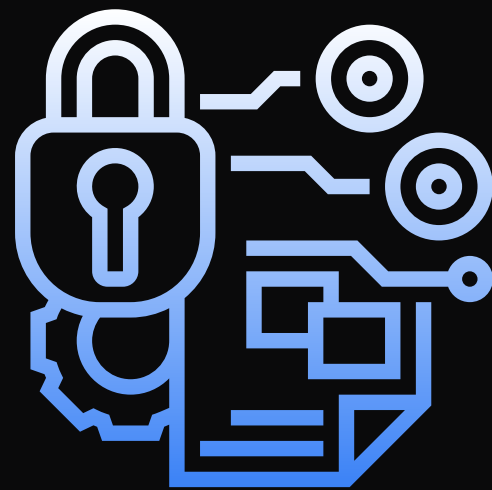




ORCHESTRATION SECURITY CENTER

FOR Small and Medium Enterprises (SMEs)

PROJECT DEVELOPMENT TEAM

Omar Abdelaziz

Team Leader & DevOps Architect

Reem Ameen

Backend System Architect

Mohamed Shabaan

Orchestration & Automation
Engineer

Youssef Abdelhady

AI Security Intelligence Engineer

Mariz Ehab

SIEM & Threat Monitoring
Specialist

Shahd Baher

Security Scanning & Vulnerability
Assessment Engineer

Omnia Helmy

Data Visualization & Network
Topology Specialist

Rahma Ibrahim

Frontend Architecture

Youssef Ali

QA & API Integration Engineer

Mazen Alaa

E2E Testing Engineer

Omar Tarek

Cyber Risk Architect & Report
Engineer

UNDER SUPERVISOR OF
DR. AMR MUHMMAD ABD-ELWAHED

SME SECURITY PROBLEM & STATS

CRITICAL CYBERSECURITY CHALLENGES IN SMES:

- Severe Expertise Gap (SMEs)
- Prohibitive Enterprise Costs
- Fragmented Security Tools
- Overwhelming Alert Fatigue
- Lack of Plain-Language Fixes

88%

of SMB breaches involved ransomware attacks

2025 VERIZON DATA BREACH INVESTIGATIONS REPORT

69%

of organizations reported incidents caused by cybersecurity skills shortages

ISC2 CYBERSECURITY WORKFORCE STUDY 2025

4.8M

global shortage of cybersecurity professionals

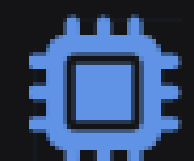
ISC2 CYBERSECURITY WORKFORCE RESEARCH 2025

OUR SOLUTION: AN INTEGRATED ORCHESTRATOR

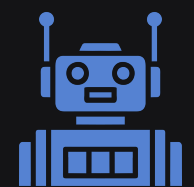
OSC CORE PLATFORM DESIGN SECTIONS:



AUTOMATED DAST PLATFORM



AGENT RECON MACHINERY



Gemini AI Advisory



Deterministic Risk Engine



Zero-Licensing Cost

SECURITY CONSOLE HUD OVERVIEW

HEALTH INDEX

30 / 100



VULNS DETECTED

18 Vulnerabilities



ACTIVE NETWORK NODES

16 Sandbox Targets



AI REPORTING STATE

PIPELINE ACTIVE



FOUR CORE

01

UNIFIED CVSS RISK ENGINE

Calculates exact CVSS weighted-quantitative risk scores (0-100) and health scores based on asset criticality.

02

GEMINI AI ADVISORY FLUIDS

Translates complex exploits into simple, plain-language business impact and step-by-step mitigation plans.

03

INTERACTIVE COGNITIVE TOPOLOGY

Visualizes attack surfaces and live scan progression through a real-time React-Force-Graph dashboard.

04

FULL REPORTING

Summarize the findings into a comprehensive report covering both the technical and business aspects.

AUTOMATION & ORCHESTRATION

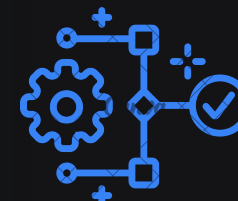


Automation

Focuses on making a single, repetitive task run on its own without manual effort. It operates in isolation, oblivious to wider system states



Executing a single standalone script or helper to make nmap to run scan in the most known 100 port



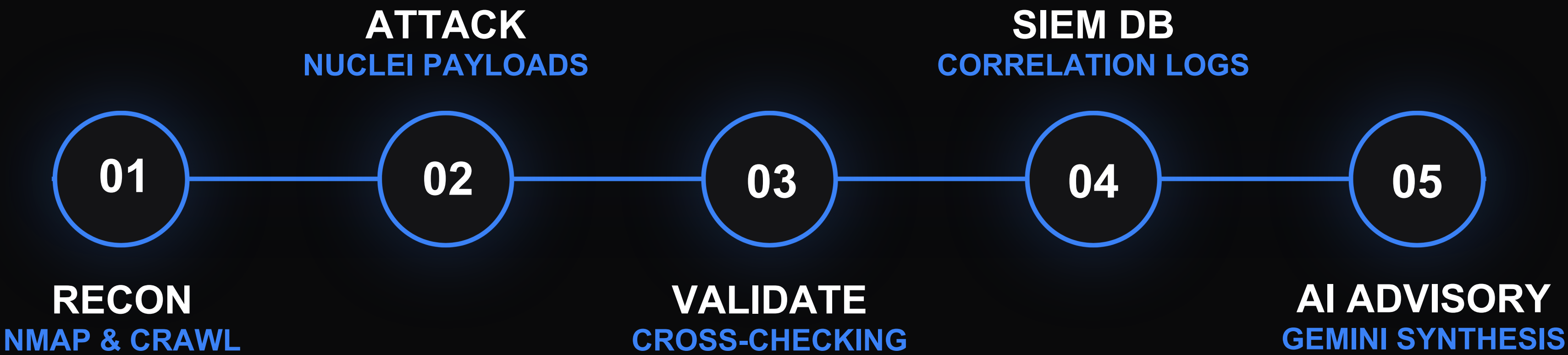
Orchestration

Coordinates multiple automated tasks across diverse environments, ensuring sequential processing, state alignment, and resilience.



Creating a pipeline from security tools to make them work in a logical sequence.

THE AUTOMATED EXPLOIT DETECTION PIPELINE



SYSTEM ARCHITECTURE DESIGN & INTEGRATIONS

CHANNELS UI & PIPELINE ASSISTANCE STACK



Frontend UI
REACT, VITE, ZUSTAND



Security Stack
NMAP, NUCLEI, OPENVAS, Wazuh



Data & Cache
POSTGRESQL, REDIS, CELERY

BACKEND APPLICATION CORE & PROCESSING



Backend Core
Python 3.11, FastAPI

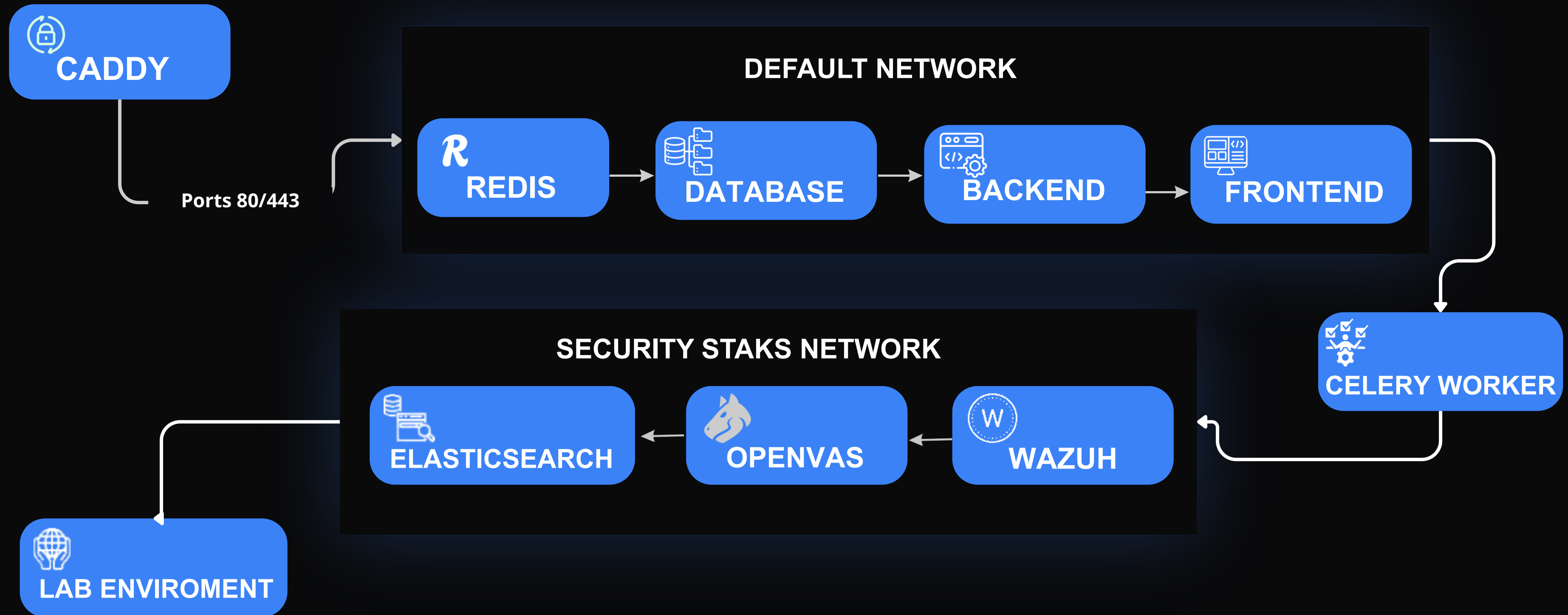


AI Engine
Google Gemini 2.0



Infrastructure
Docker Compose Orchestration

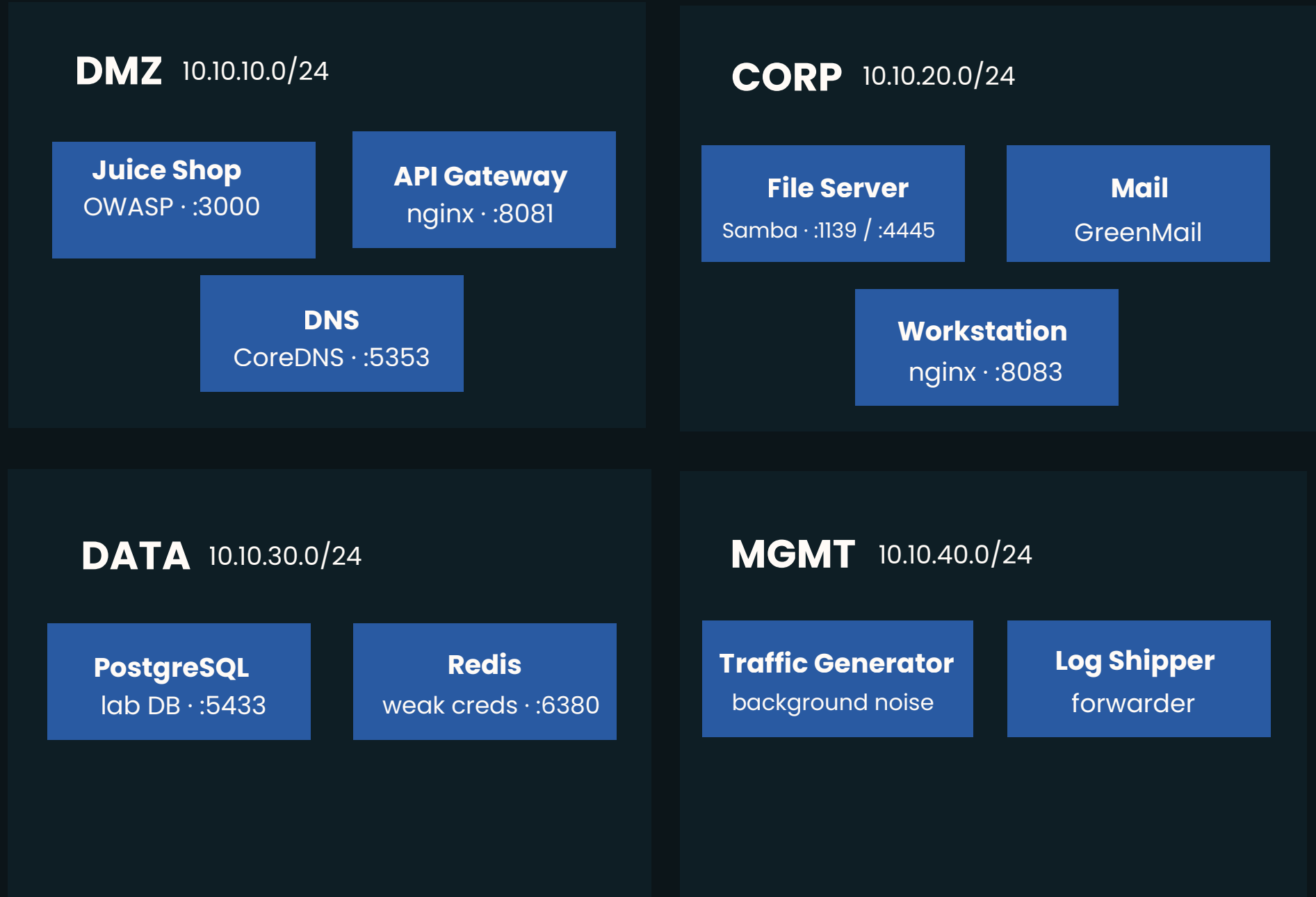
MICROSERVICES ARCHITECTURE



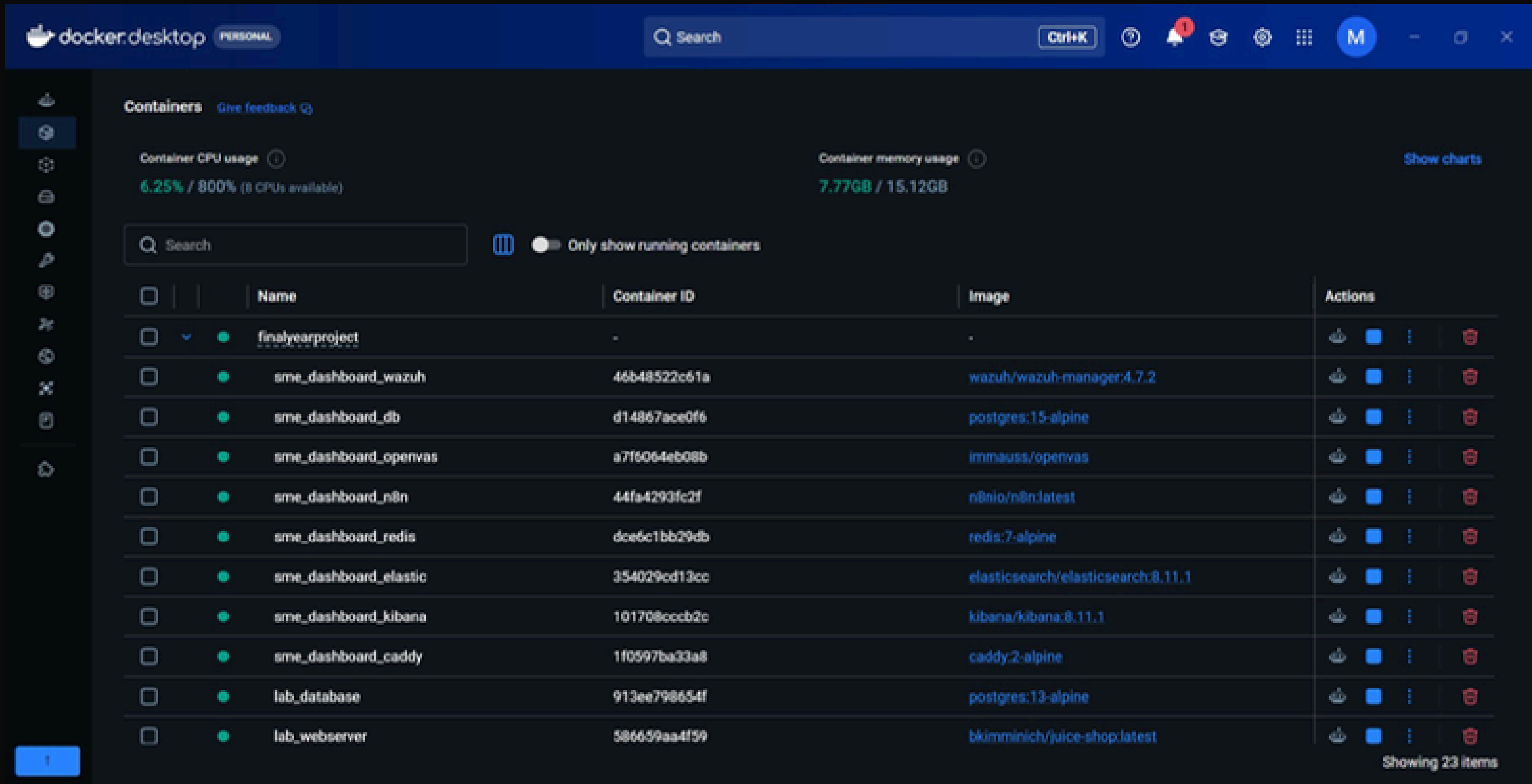
LAB & TEST ENVIROMENT

- SME Simulation Environment
- Real Vulnerabilities
- Simulated Subnets
- Traffic & Log Generator

Lab Environment



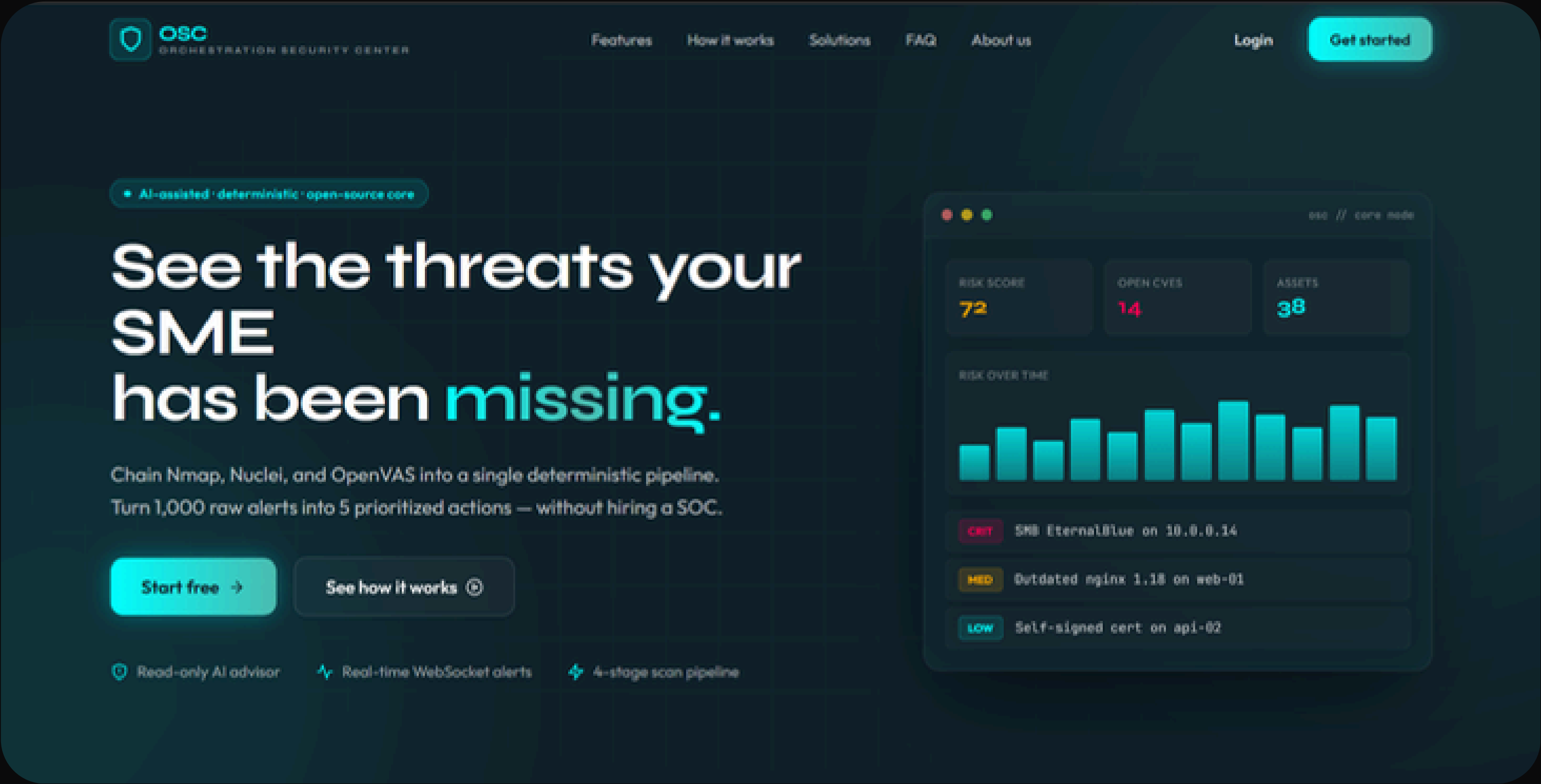
INFRASTRUCTURE DOCKER COMPOSE ORCHESTRATION



INTERACTIVE DEMO

PRACTICAL IMPLEMENTATION

HOME PAGE



SIGNUP PAGE

← BACK TO HOME

OSC

CREATE YOUR ACCOUNT

NEW ACCOUNT REGISTRATION

EMAIL

you@example.com

FULL NAME (OPTIONAL)

Jane Doe

PASSWORD

At least 8 characters

CONFIRM PASSWORD

Re-enter password

CREATE ACCOUNT

Already have an account? [Sign in](#)

New accounts start with VIEWER role. Contact your admin to upgrade.

LOGIN PAGE

← BACK TO HOME

OSC

ORCHESTRATION SECURITY CENTER

SECURE AUTHENTICATION

USERNAME / EMAIL

admin@local

PASSWORD

••••••••

ACCESS SYSTEM

Need an account?

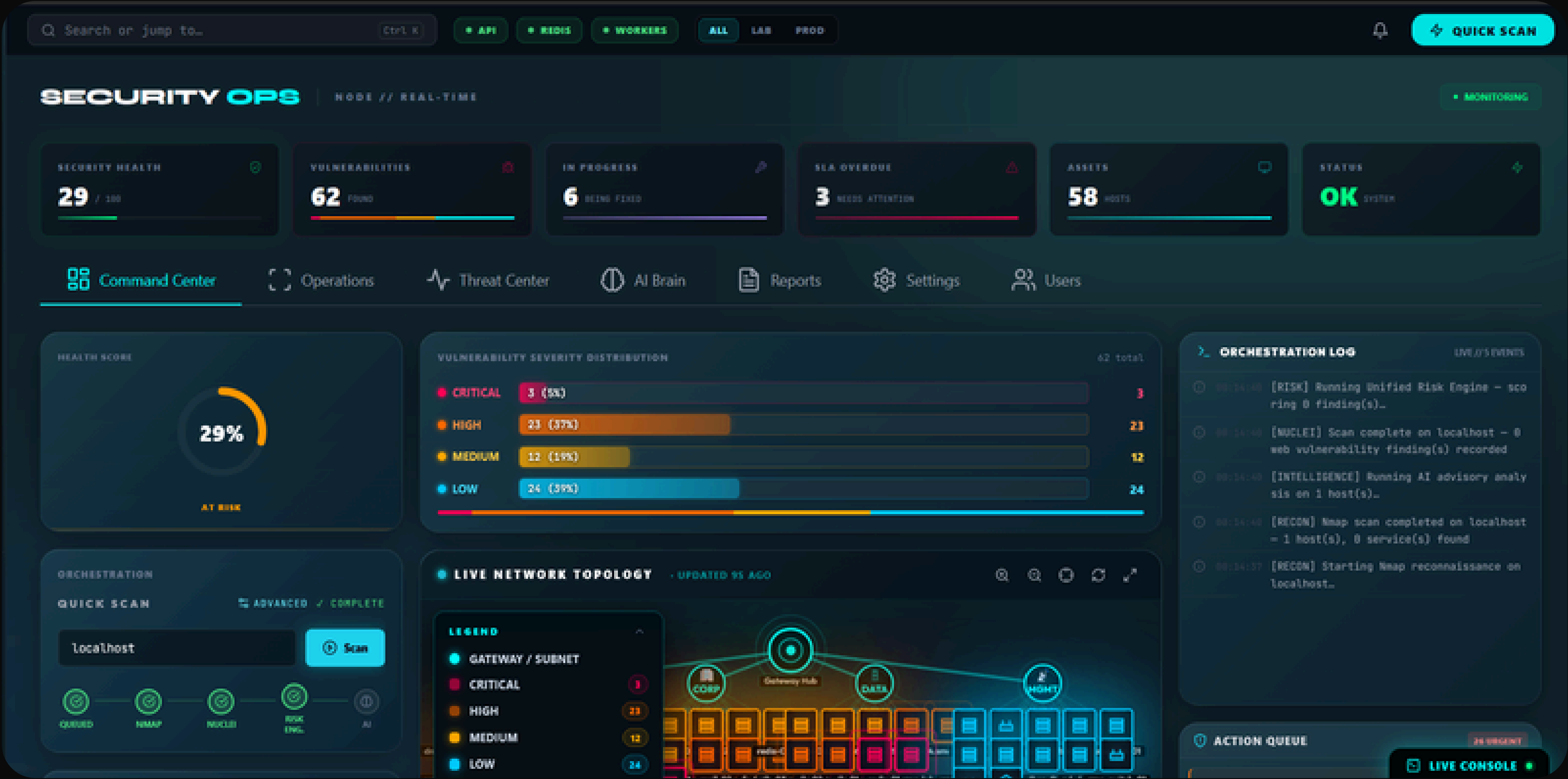
Create one

AUTHORIZED PERSONNEL ONLY

All actions are logged and monitored

ADMIN PRIVILEGE

1- COMMAND CENTER



1- COMMAND CENTER

1.1 NAVIGATION BAR ICONS

 **QUICK SCAN**

 **CONFIRM SCAN**

TARGET

DATA 10.10.30.0/24

+5 others

PROFILE

Quick

nmap top-1000 ports

ETA

~2 min

 **RUN**

 **CONFIGURE**

Enter

 to run ·

C

 to configure ·

Esc

 to cancel



 **NOTIFICATIONS** all read ✓ **MARK READ** ✕



Scan Complete — 0 finding(s) on localhost 3h ago

3 critical · 40 high · 2 medium · 33 low
· 65 info
localhost



Scan Started — localhost 3h ago

Security scan initiated on localhost.
Running: Nmap recon → Nuclei web scan →...
localhost

2 events · live from WebSocket

2-OPERATION

2.1 SCAN

SCANHISTORYNODESLAB

QUICK SCAN

ADVANCED

localhost

Scan

QUICK

MANUAL

RECURRING

RISK ANAL

AI

AUTOMATED SCANNING

TARGET

172.30.0.20

FREQUENCY

Every day

TIME

02:00 AM

STATUS 0 2 0 0 0

SCHEDULE

View Schedule

Scan fires automatically at the configured time.

VULNERABILITY RISK DISTRIBUTION

77 total

CRITICAL

4%

3

HIGH

52%

49

MEDIUM

1%

1

LOW

43%

33

3 CRITICAL

49 HIGH

1 MEDIUM

33 LOW

Sample findings — kick off a scan to populate this panel with live results.

SQL Injection in /api/login

10.10.10.10 - CRITICAL

Outdated nginx 1.18 (CVE-2021-23017)

10.10.10.20 - HIGH

Weak TLS Cipher Suites

10.10.20.10 - MEDIUM

LIVE CONSOLE

2-OPERATION

2.1 SCAN

 **AUTOMATED SCANNING**

TARGET

172.30.0.20

FREQUENCY

Every day

TIME

🕒 02:00 AM

cron: 0 2 * * *

SAVE SCHEDULE

Hide schedules

172.30.0.20

0 2 * * *



Scan fires automatically at the configured time.



SQL Injection in /api/login

10.10.10.10 • CRITICAL

DESCRIPTION

Unsanitised user input flows into a parameterised SQL query, allowing authentication bypass.

REMEDiation

Use parameterised queries / prepared statements. Validate all input server-side.

2.1 SCAN

2.1.1 ADVANCED SCAN

NEW SCAN

1 TARGET

>

2 PROFILE

>

3 SCHEDULE

>

4 REVIEW

Target

Who are we scanning?

Existing target

Manual URL / IP

Lab Environment

Select a target...

Select a target...

[Lab] Corporate API Gateway — http://lab_api_gateway:8081

[Lab] E-Commerce Web Server — http://lab_webserver:3000

MGMT 10.10.40.0/24 — 10.10.40.0/24

DATA 10.10.30.0/24 — 10.10.30.0/24

CORP 10.10.20.0/24 — 10.10.20.0/24

DMZ 10.10.10.0/24 — 10.10.10.0/24

2.1 SCAN

2.1.1 ADVANCED SCAN

 NEW SCAN

1 TARGET

>

2 PROFILE

>

3 SCHEDULE

>

4 REVIEW

 **Target** – Who are we scanning?

Existing target

Manual URL / IP

 Lab Environment

127.0.0.1

Looks valid. Scope enforcement will consider this outbound.

STANDARD • 2 tools

Cancel

Continue

2.1 SCAN

2.1.1 ADVANCED SCAN (LAB ENVIROMENT)

 NEW SCAN

1 TARGET

>

2 PROFILE

>

3 SCHEDULE

>

4 REVIEW

 **Target** — Who are we scanning?

Existing target

Manual URL / IP

 Lab Environment

One scan per subnet — every selected zone becomes its own History row.

DESELECT ALL

☒ **DMZ**
Internet-facing services

10.10.10.0/24

☒ **CORP**
Corporate office tier

10.10.20.0/24

☒ **DATA**
Database & cache tier

10.10.30.0/24

☒ **MGMT**
Management & monitoring

10.10.40.0/24

Launches 4 parallel scans — one per subnet.

STANDARD · 2 tools

Cancel

Continue

2.1 SCAN

2.1.1 ADVANCED SCAN (LAB ENVIROMENT)

NEW SCAN

TARGET

2 PROFILE

3 SCHEDULE

4 REVIEW

Profile

How deep?

QUICK

~2 min

Top-1000 port sweep. Use for liveness / triage.

NMAP

STANDARD

~10 min

Port sweep + Nuclei template checks. The daily driver.

NMAPNUCLEI

FULL

~20 min

Standard plus AI-powered validation of each finding.

NMAPNUCLEIAI_VALIDATION

ADVANCED

Custom tools · Reporting · SIEM

STANDARD · 2 tools

BackCancelContinue

ADVANCED

Custom tools · Reporting · SIEM

TOOLS

Selecting tools individually switches the profile to Custom.

Nmap

Port discovery & service detection.

☒

Nuclei

Template-based vulnerability checks.

☒

OpenVAS

Authenticated vulnerability scanner.

☐

AI Validation

LLM agent re-checks each finding to suppress false positives.

☐

POST-SCAN ACTIONS

Auto-generate PDF report

☐

Forward findings to SIEM

☐

STANDARD · 2 tools

BackCancelContinue

2.1 SCAN

2.1.1 ADVANCED SCAN (LAB ENVIROMENT)

NEW SCAN

TARGET

PROFILE

3 SCHEDULE

4 REVIEW

Schedule

Now or recurring?

Run once (now)

Every day at 02:00

Every Monday 02:00

Every hour

Custom cron...

STANDARD · 2 tools

Back

Cancel

Continue

NEW SCAN

TARGET

PROFILE

SCHEDULE

4 REVIEW

Review

Confirm & launch.

TARGET

127.0.0.1

PROFILE

STANDARD

NMAP

NUCLEI

SCHEDULE

Run once, now.

POST-SCAN

PDF report: yes

SIEM forward: no

EXACT REQUEST PAYLOAD

STANDARD · 2 tools

Back

Cancel

Launch Scan

2-OPERATION

2.2 HISTORY

Command Center

Operations

Threat Center

AI Brain

Reports

Settings

Users

SCAN

HISTORY

NODES

LAB

SCAN HISTORY 63 total

Target...

Any status

Any profile

mm / dd / yyyy

mm / dd / yyyy

Compact

Refresh

TIME	TARGET	PROFILE	STATUS	FINDINGS	DURATION	ACTIONS
> 11m ago	localhost	QUICK	Completed	INFO 0	1m 38s	EXPLAIN REPORT
> 19m ago	localhost	QUICK	Completed	INFO 0	4s	EXPLAIN REPORT
> 21m ago	localhost	QUICK	Completed	INFO 0	5s	EXPLAIN REPORT
> 1h ago	localhost	QUICK	Completed	INFO 0	6m 45s	EXPLAIN REPORT
> 1h ago	127.0.0.1	STANDARD	Completed	INFO 0	8m 27s	EXPLAIN REPORT
> 1h ago	localhost	QUICK	Completed	INFO 0	5s	EXPLAIN REPORT
> 2h ago	localhost	QUICK	Completed	INFO 0	3m 54s	EXPLAIN REPORT
> 2h ago	localhost	QUICK	Completed	INFO 0	10s	EXPLAIN REPORT
> 3h ago	localhost	QUICK	Completed	INFO 0	1m 41s	EXPLAIN REPORT

LIVE CONSOLE

2-OPERATION

2.2 Deep Dive History

4h ago

10.10.30.0/24

QUICK

Completed

CRITICAL

7

20s

EXPLAIN

REPORT

SCAN SUMMARY

ID	9596aa9f-3f8b-4335-8f65-278ba293963d	RISK SCORE	31.5
TARGET	10.10.30.0/24	VULNERABILITIES	7
PROFILE	quick	ASSETS	1
STARTED	5/25/2026, 7:11:41 AM	AUTO-REPORT	off
COMPLETED	5/25/2026, 7:12:01 AM	SIEM FORWARD	off
DURATION	20s		

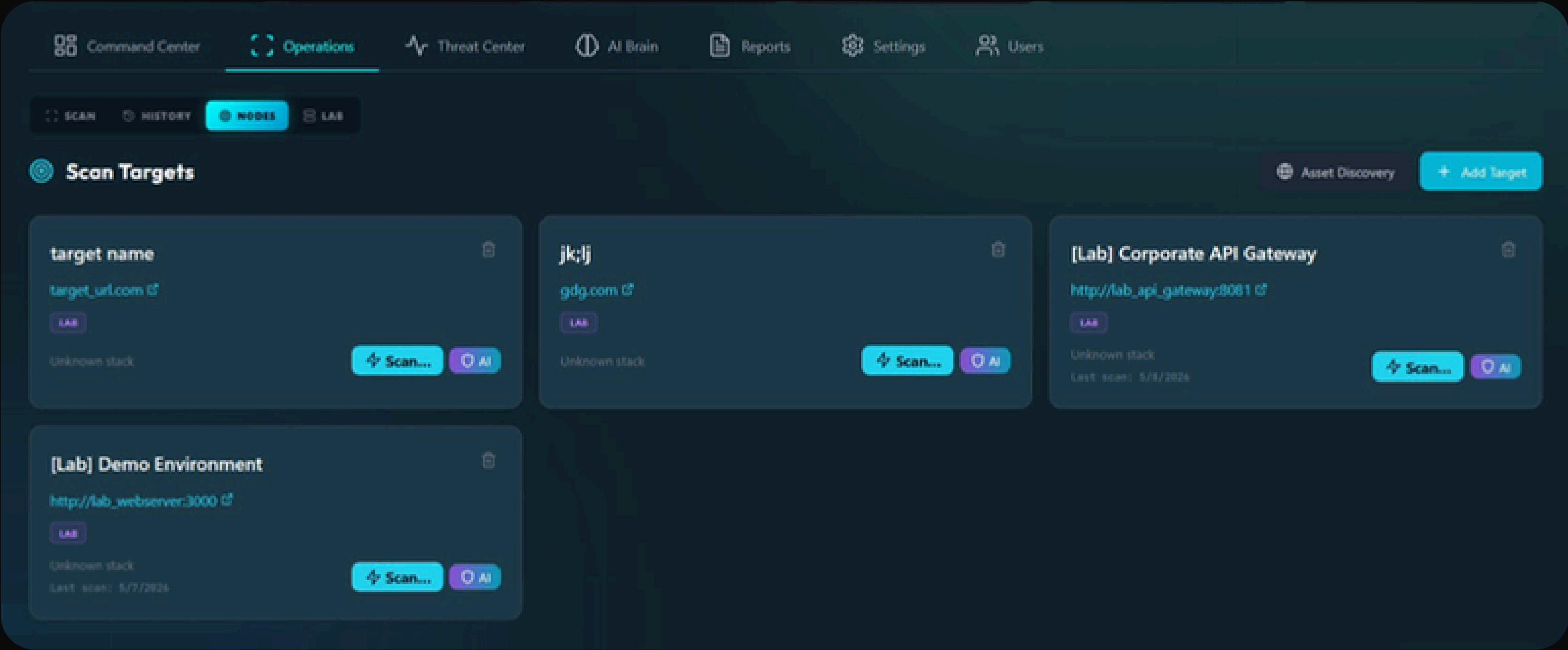
FINDINGS

COMMAND PREVIEW

```
POST /api/v1/scans/
{
  "target_url": "10.10.30.0/24",
  "scan_type": "quick",
  "tools": [
    "nmap"
  ],
  "auto_report": false,
  "siem_forward": false
}
```


2-OPERATION

2.3 NODES



2.3 NODES

2.3.1 ASSET DISCOVERY

SCANHISTORYNODESLAB

Scan Targets

Asset DiscoveryAdd Target

Automated Asset Discovery

Enter a root domain to automatically find subdomains and add them as targets using Subfinder.

hackerone.comStart Discovery

Discovery Complete!VIA CRT.SH

Found 15 subdomains.
Created 15 new targets.

NEW TARGETS ADDED:

api.hackerone.com

design.hackerone.com

docs.hackerone.com

events.hackerone.com

go.hackerone.com

gslink.hackerone.com

hackerone.com

info.hackerone.com

links.hackerone.com

mta-sts.forwarding.hackerone.com

mta-sts.hackerone.com

mta-sts.managed.hackerone.com

support.hackerone.com

websockets.hackerone.com

www.hackerone.com

2.3 NODES

2.3.2 ADD TARGET

 ADD ENVIRONMENT – IDENTITY

1 IDENTITY

2 ENVIRONMENT

3 COMPLIANCE

4 SAFETY

5 REVIEW

NAME *

localhost

BASE URL / IP *

127.0.0.1

NOTES

Owner team, change window, contact email, known caveats...

← Back

Next →

 ADD ENVIRONMENT – ENVIRONMENT

1 IDENTITY

2 ENVIRONMENT

3 COMPLIANCE

4 SAFETY

5 REVIEW

ENVIRONMENT TYPE

LAB

Internal test bed; full aggressive scans allowed

DEVELOPMENT

Dev stage; full scans but rate-limited

STAGING

Pre-prod mirror; safe-mode scans

PRODUCTION

Live traffic; safe, low-RPS scans only

ASSET VALUE

LOW

MEDIUM

HIGH

CRITICAL

DATA SENSITIVITY

NONE

PII

FINANCIAL

← Back

Next →

2.3 NODES

2.3.2 ADD TARGET

ADD ENVIRONMENT – COMPLIANCE

1 IDENTITY

2 ENVIRONMENT

3 COMPLIANCE

4 SAFETY

5 REVIEW

COMPLIANCE FRAMEWORKS (FINDINGS WILL BE MAPPED TO THESE)

☐

PCI-DSS

☐

HIPAA

☐

ISO-27001

☐

GDPR

☐

NIST CSF

☐

OWASP ASVS

← Back

Next →

ADD ENVIRONMENT – SAFETY

1 IDENTITY

2 ENVIRONMENT

3 COMPLIANCE

4 SAFETY

5 REVIEW

MAX REQUESTS PER SECOND

10

Production targets should stay under 10. Lab targets can go up to ~100.

SCOPE ALLOWLIST (HOSTS/CIDRS THE SCANNER MAY TOUCH)

example.com or 10.0.0.0/24

+ Add

Empty ⇒ only the base URL's hostname will be allowed.

2.3 NODES

2.3.2 ADD TARGET

✓ ADD ENVIRONMENT – REVIEW

1 IDENTITY

2 ENVIRONMENT

3 COMPLIANCE

4 SAFETY

5 REVIEW

NAMElocalhost

BASE URL127.0.0.1

ENVIRONMENTlab

ASSET VALUEMEDIUM

DATA SENSITIVITYNONE

COMPLIANCE TAGS–

MAX RPS10

SCOPE ALLOWLIST(base hostname only)

NOTES–

← Back

Create Target

2-OPERATION

2.4 LAB

Living Lab Environment

HEALTHY

8

Containers Running

108467

Events

45257

Alerts

high

Traffic

Seed Targets

DMZ Zone (3 services)

E-Commerce Web Server

Running

DMZ

OWASP Juice Shop — intentionally vulnerable web application

sqli

xss

bofa

+3

Port: 3000

CVSS: 9.5

lab_webserver

Scan

Corporate API Gateway

Running

DMZ

API gateway with information disclosure vulnerabilities

info-disclosure

header-leak

directory-listing

+1

Port: 8081

CVSS: 6

lab_api_gateway

Scan

Corporate DNS Server

Running

DMZ

CoreDNS server with zone transfer enabled

zone-transfer

recursion-enabled

CORP Zone (2 services)

Corporate File Server

Running

CORP

Samba file server with weak credentials and exposed shares

weak-credentials

smb-enum

sensitive-data-exposure

Port: 445

CVSS: 8

lab_fileserver

Corporate Mail Server

Running

CORP

Greenmail server with plaintext SMTP/POP3/IMAP

weak-credentials

plaintext-protocols

Port: 3025

CVSS: 7

lab_mailserver

LIVE CONSOLE

2-OPERATION

2.4 LAB

DATA Zone (2 services)

Production Database

Running

DATA

PostgreSQL with weak password and sensitive data in plaintext

weak-credentials

pii-plaintext

no-encryption

Port: 5432CVSS: 9lab_database

Redis Cache

Running

DATA

Redis with no authentication and protected-mode disabled

no-auth

unauthenticated-access

data-exfiltration

Port: 6380CVSS: 8.5lab_redis_cache

MGMT Zone (1 services)

Hardened Bastion

Running

MGMT

Hardened bastion host — patched, MFA-enforced, no exposed vulnerabilities

Port: 2222CVSS: 0lab_bastion

Live Event Feed

18:53:00	web	http_request	success
18:52:59	suspicious	bulk_data_query	success
18:52:58	dns	lookup	success
18:52:58	database	query	success
18:52:57	cache	redis_info	success
18:52:56	web	http_request	success
18:52:55	authentication	smb_access	failure
18:52:53	dns	lookup	success
18:52:53	database	query	success

3-THREAT CENTER

3.1 SIEM

SIEMVULNS

TOPOLOGY

Unified SIEM Inbox

Live

+3 new

Updated 1:54:01 AM

Pause

All 181

Critical 10

High 50

Medium 31

Low 90

Search alerts — rule, description, agent, IP, MITRE, full log...

MEDIUM • Lvl6

5/26/2026, 1:53:57 AM

Web application error

lab-monitor10.10.20.51webaccesslog

HIGH • Lvl10

5/26/2026, 1:53:55 AM

Authentication failure

lab-monitor10.10.20.40authentication_failurehigh

LOW • Lvl3

5/26/2026, 1:53:50 AM

Lab event: database/query

lab-monitor10.10.20.40ome-lablow

LOW • Lvl3

5/26/2026, 1:53:46 AM

Cache operation

lab-monitor10.10.10.10cachelow

LIVE CONSOLE

3.1 SIEM

3.1.1 SIEM ALERT DEEP DIVE

MEDIUM · Level 6

Rule 31101

5/26/2026, 1:54:10 AM

×

Web application error

web

accesslog

medium

RULE DETAILS

Rule ID	31101
Level	6 (MEDIUM)
Description	Web application error
Groups	web, accesslog, medium

AGENT / SOURCE

Agent	lab-monitor
Agent ID	100
Agent IP	10.10.40.30
Location	lab/dmz/web
Source IP	10.10.20.51
Dest IP	10.10.10.10

FULL LOG

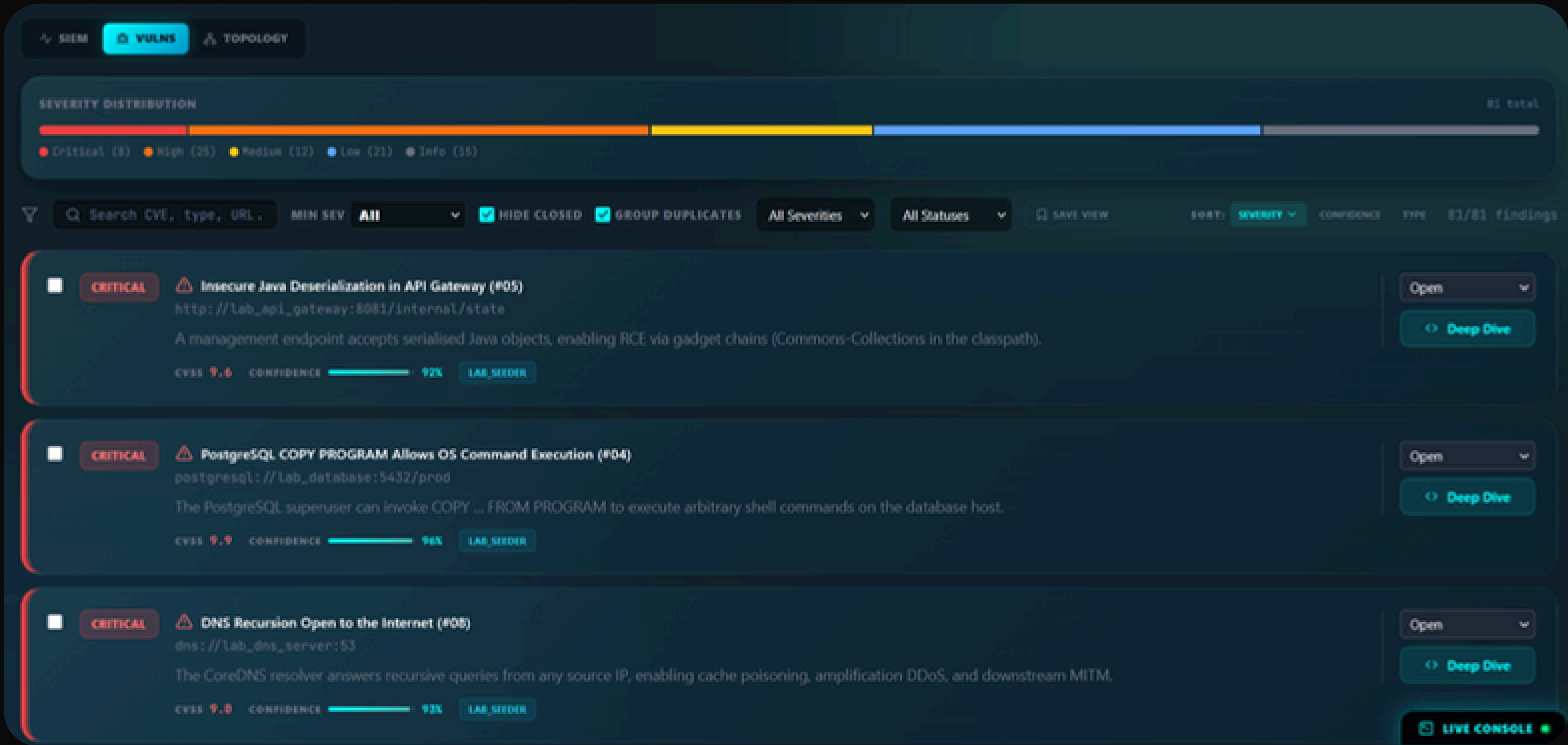
```
{"@timestamp": "2026-05-25T22:54:10.500816+00:00", "event_category": "web", "event_action": "http_request", "source_ip": "10.10.20.51", "destination_ip": "10.10.10.10", "destination_port": 3000, "protocol": "HTTP", "status": "failure", "severity": "medium", "details": {"method": "GET", "path": "..."}}
```

Copy JSON

Close

3-THREAT CENTER

3.2 VULNERABILITIES



3-THREAT CENTER

3.2 VULNERABILITIES

CRITICAL

 PostgreSQL COPY PROGRAM Allows OS Command Execution (#04)

postgresql: //lab_database:5432/prod

The PostgreSQL superuser can invoke COPY ... FROM PROGRAM to execute arbitrary shell commands on the database host.

CVEs 9.9

CONFIDENCE 96%

LAB_MITIG

In Progress

Open

In Progress

Fixed

False Positive

Accepted (Risk)

3.2 VULNERABILITIES

3.2.1 VULNERABILITIES DEEP DIVE

CRITICAL

< 1 / 81 > X

Insecure Java Deserialization in API Gateway (#05)

http://lab_api_gateway:8081/internal/state
10.10.10.13:8081 (http)

CVSS SCORE

9.6 / 10

CVSS:3.1/AV:N/AC:L/PR:N/UI:N...

CONFIDENCE

92%

STATUS

OPEN

DETECTED BY

lab_seeder

DESCRIPTION

A management endpoint accepts serialised Java objects, enabling RCE via gadget chains (Commons-Collections in the classpath).

CVSS BREAKDOWN

V3.X

Attack Vector

Network

Attack Complexity

Low

Privileges

None

User Interaction

None

Scope

Unchanged

Confidentiality

High

Integrity

High

Availability

High

AI SECURITY BRIEF

GEMINI AI

ATTACK SCENARIO

An attacker probing http://lab_api_gateway:8081/internal/state would identify this Insecure Java Deserialization in API Gateway (#05) exposure through automated scanning or manual reconnaissance. The severitylevel.critical-severity nature makes it an attractive entry point for gaining unauthorized access or escalating privileges within the network.

AI SECURITY BRIEF

GEMINI AI

ATTACK SCENARIO

An attacker probing http://lab_api_gateway:8081/internal/state would identify this Insecure Java Deserialization in API Gateway (#05) exposure through automated scanning or manual reconnaissance. The severitylevel.critical-severity nature makes it an attractive entry point for gaining unauthorized access or escalating privileges within the network.

BUSINESS IMPACT

Successful exploitation could lead to unauthorized data access, service disruption, or full system compromise. Depending on the asset's role, this may trigger compliance violations (PCI DSS, GDPR) and result in reputational and financial damage.

REMEDIALATION STEPS

1

Immediately restrict network-level access to the affected service using firewall rules

2

Apply the latest vendor security patch or update the software to a non-vulnerable version

3

Harden the service configuration (disable unnecessary features, enforce authentication)

4

Rotate any credentials, tokens, or API keys that may have been exposed

5

Re-run a targeted scan against the affected endpoint to confirm remediation

DETECTION & MONITORING

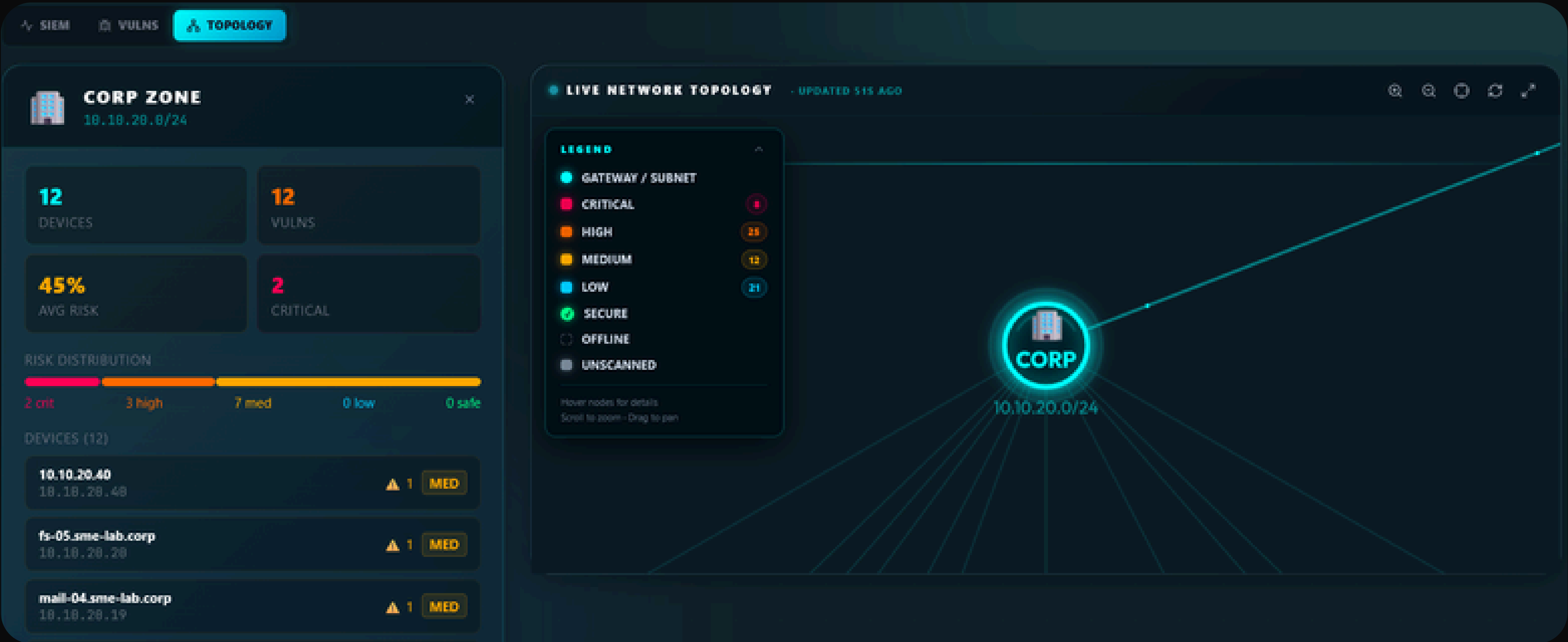
Monitor application and system logs for anomalous access patterns or error spikes consistent with exploitation attempts. Configure SIEM rules to alert on indicators of compromise (unusual user-agents, payloads, or response codes) for this service.

VERIFY THE FIX

Re-execute the original scanner or proof-of-concept against the patched endpoint and confirm the vulnerability no longer reproduces. Validate service functionality under normal operations to rule out regressions.

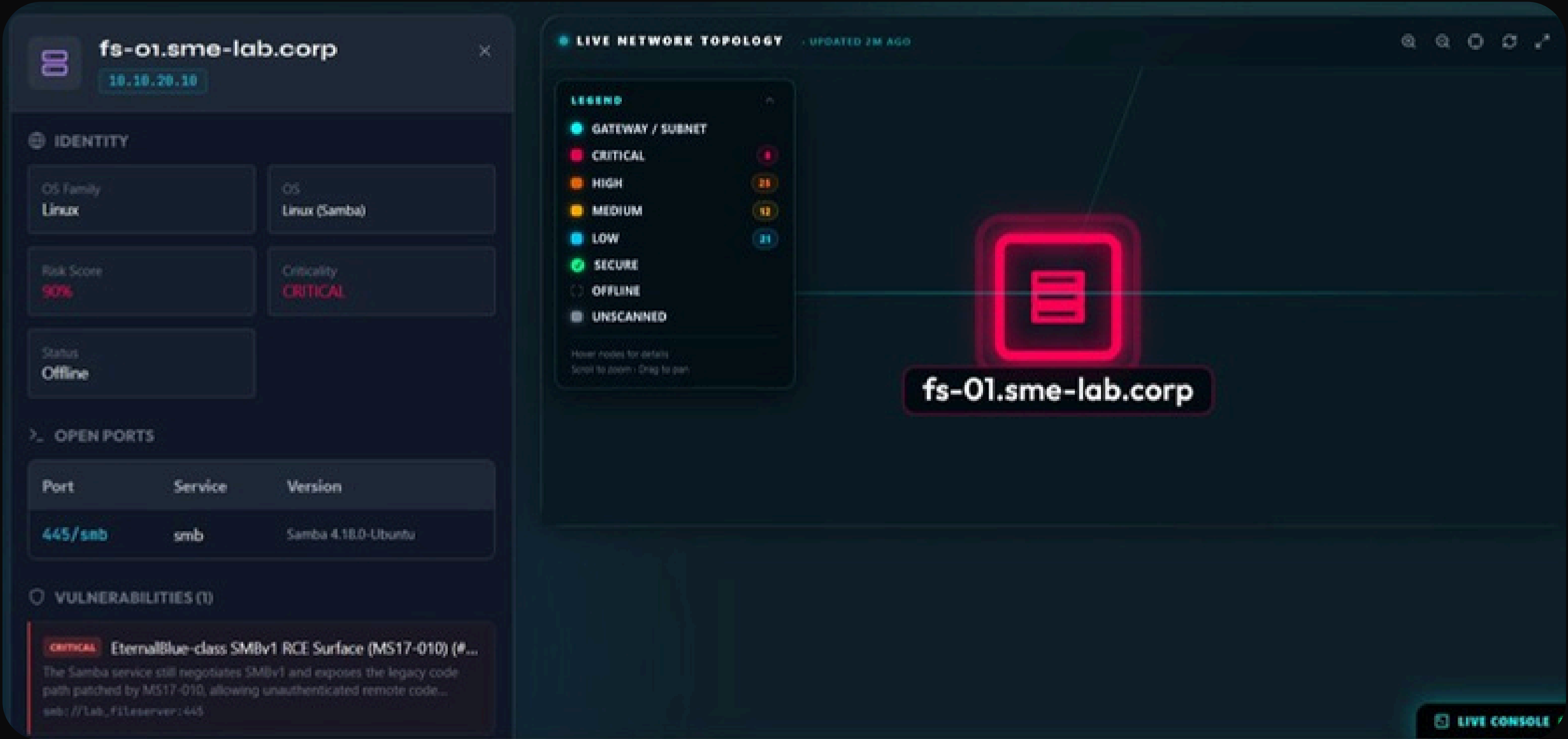
3-THREAT CENTER

3.3 TOPOLOGY



3-THREAT CENTER

3.3 TOPOLOGY



4- AI BRAIN

Command Center

Operations

Threat Center

AI Brain

Reports

Settings

Users

SECURITY PIPELINE

NMAP

HUCLER

RISK ENGINE

AI ADVISORY

NEURAL CONSOLE

STREAM ACTIVE (4 DATA POINTS)

SYNC SCROLL

[RECON]

Network Reconnaissance Completed

22:30:43

[ATTACK]

Vulnerability Scan Completed

22:31:49

[VALIDATION]

Risk Scoring Completed

22:31:49

[REPORTING]

AI Advisory Report Generated

22:31:50

LINK ESTABLISHED

ENCRYPTION: AES-256

AI AGENT MONITOR V1.2.0-PRO

5- REPORT

Command Center

Operations

Threat Center

AI Brain

Reports

Settings

Users

Completed Scans

10.10.20.0/24

5/25/2026

Completed Risk 35.2

10.10.10.0/24

5/25/2026

Completed Risk 31.5

10.10.30.0/24

5/25/2026

Completed Risk 31.5

10.10.40.0/24

5/25/2026

Completed Risk 31.5

10.10.30.0/24

5/25/2026

Completed Risk 28.3

10.10.20.0/24

Completed: 5/25/2026, 10:31:50 PM

Generate Report

Full Report

VULNERABILITY SUMMARY

SELECTED SCAN ONLY

0

Critical

2

High

0

Medium

3

Low

Click **Generate Report** to build a signed PDF report for this scan.

5- REPORT (SELECTED SCAN REPORT)

10.10.20.0/24

Completed: 5/25/2026, 11:17:30 PM

Export PDF

Generate Report

Full Report

VULNERABILITY SUMMARY

SELECTED SCAN ONLY

0
Critical

3
High

0
Medium

3
Low

Report generated successfully

Report ID

Generated

Digitally Signed

Findings Hash

d5e630a5-152a-4e91-b0c6-41895c1ed15c

5/25/2026, 11:17:44 PM

Ver

a41244d270e59ce7c5c30a37aee2e80d170dfb39b29b64d836

LIVE CONSOLE

5- REPORT (FULL REPORT)

localhost

Completed: 5/27/2026, 9:14:32 PM

Export PDF

Generate Report

Full Report

VULNERABILITY SUMMARY

FULL REPORT SCOPE · ALL SCANS

3
Critical

23
High

12
Medium

24
Low

Report generated successfully

Report ID

90a9805e-4c42-45ca-84b9-1bd846681ea8

Generated

5/27/2026, 9:17:12 PM

Digitally Signed

Yes

Findings Hash

9a3b8f0c7e771ed12fff066fd901f1225fc19cdd440997209e25e17867b0f332

Click **Export PDF** above to download the signed report.

FINDINGS (83)

6-SETTING

☰ Command Center

⌕ Operations

📶 Threat Center

🧠 AI Brain

📄 Reports

⚙️ Settings

👤 Users

⚙️

TOOL TOGGLES
Enable or disable scan tools and integrations at runtime

🔍

OpenVAS Scanner
Network vulnerability scanner (OVM/GMP)

🔛

🧠

LLM AI Validation
Gemini triage of findings (never overrides reprobe)

🔛

🛡️

Nuclei Scanning
Template-driven vulnerability scanner

🔛

📶

SIEM Integration
Elasticsearch + Wazuh event pipeline

🔛

🔍

Nmap Scanning
Port and service discovery

🔛

🧠

SCAN & LLM DEFAULTS
Per-scan and per-day limits enforced by the orchestrator

DAILY LLM TOKEN BUDGET

500000

🔄

PER-SCAN LLM TOKEN CAP

50000

🔄

6-SETTING

 **LAB ENVIRONMENT CONTROLS**
Living lab container lifecycle

CONTAINERS

7 RUNNING

OVERALL

HEALTHY

TRAFFIC INTENSITY

high

Seed Targets

Refresh Status

 **INTEGRATION HEALTH**
Live reachability probe of every configured service

ELASTICSEARCH HTTP 200	CONNECTED	HAZUH HTTP 401	CONNECTED
REDIS PING OK	CONNECTED	POSTGRES SELECT 1 OK	CONNECTED
OPENVAS TCP openvas:9390 OK	CONNECTED	GEMINI API key present	CONFIGURED

7- USERS

Command Center

Operations

Threat Center

AI Brain

Reports

Settings

Users

USER MANAGEMENT

3 users

Refresh

+ New User

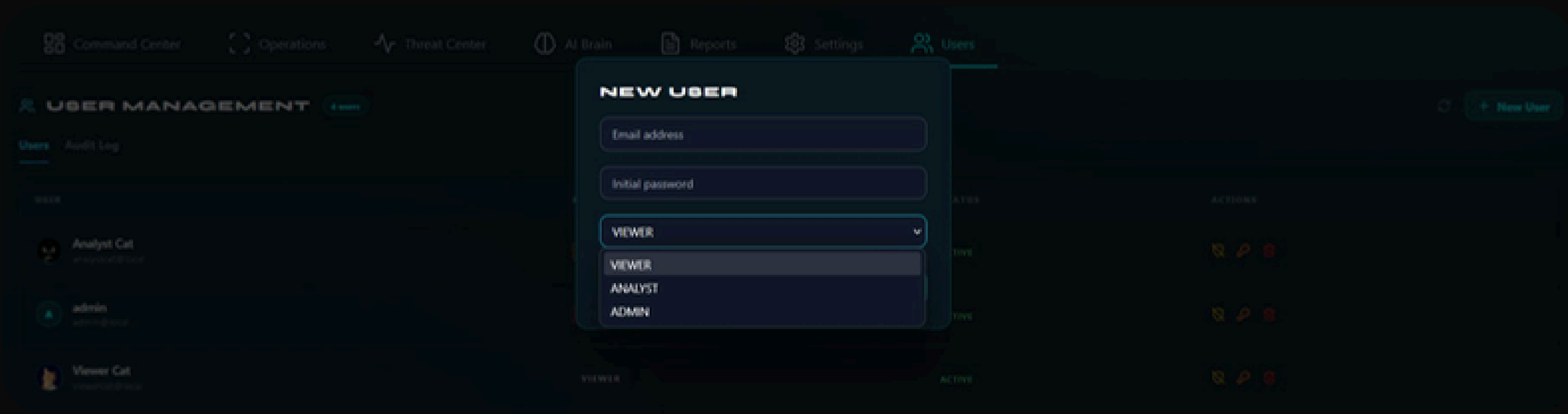
Users

Audit Log

USER	ROLE	STATUS	ACTIONS
A admin admin@local	ADMIN	ACTIVE	YOU
A analyst analyst@local MUST CHANGE PASSWORD	ANALYST	ACTIVE	Profile Deactivate Reset Password Delete
V viewer viewer@local MUST CHANGE PASSWORD	VIEWER	ACTIVE	Profile Deactivate Reset Password Delete

7- USERS

7.1 NEW USER



7- USERS

7.1 NEW USER

USER	ROLE	STATUS	ACTIONS
A admin admin@local	ADMIN	ACTIVE	YOU
A analyst analyst@local MUST CHANGE PASSWORD	ANALYST	ACTIVE	
V viewer viewer@local MUST CHANGE PASSWORD	VIEWER	ACTIVE	

CHANGE ROLE – ANALYST@LOCAL

VIEWER

ANALYST ✓

ADMIN

7- USERS

7.2 AUDIT LOG

 **USER MANAGEMENT**

4 users

+ New User

Users

Audit Log

USER_LOGIN

admin@local → admin@local

7/5/2024, 6:42:31 PM

USER_LOGIN

analyst@local → analyst@local

7/5/2024, 6:40:01 PM

USER_CREATED

admin@local → mazenataa188@gmail.com

7/5/2024, 6:39:04 PM

USER_LOGIN

admin@local → admin@local

7/5/2024, 6:38:49 PM

USER_LOGIN

viewer@local → viewer@local

7/5/2024, 6:38:34 PM

USER_LOGIN

analyst@local → analyst@local

 LIVE CONSOLE

VIEWER PRIVILEGE

VIEWER PRIVILEGE

1.SCAN DISABLED

 Command Center

 Operations

HEALTH SCORE

5%

CRITICAL

ORCHESTRATION

QUICK SCAN

ADVANCED

localhost

 Scan

ADMIN / ANALYST ONLY

 QUEUED

 NMAP

 NUCLEI

 RISK ENG.

 AI

VIEWER PRIVILEGE

2. VULNERABILITY STATUS MODIFICATION LOCKED

SIEMVULNSTOPOLOGY

SEVERITY DISTRIBUTION93 total

Critical (8)

High (28)

Medium (12)

Low (24)

Info (21)

Search CVE, type, URL.

MIN SEVAll

☒ HIDE CLOSED

☒ GROUP DUPLICATES

All Severities

All Statuses

SAVE VIEW

SORT:SEVERITY

CONFIDENCE

TYPE

85/93 findings

CRITICAL

Insecure Java Deserialization in API Gateway (#05)

http://lab_api_gateway:8081/internal/state

A management endpoint accepts serialised Java objects, enabling RCE via gadget chains (Commons-Collections in the classpath).

CVEs9.6

CONFIDENCE92%

LAB_MIDDER

In Progress

ADMIN / ANALYST ONLY

Deep Dive

VIEWER PRIVILEGE

3. FULL REPORT GENERATION ONLY

10.10.20.0/24

Completed: 5/25/2026, 11:17:30 PM

 Generate Report

 Full Report

GENERATE REPORT IS ADMIN / ANALYST ONLY – VIEWERS CAN USE FULL REPORT.

VULNERABILITY SUMMARY

SELECTED SCAN ONLY

0
Critical

3
High

0
Medium

3
Low

VIEWER PRIVILEGE

4.SETTING

🔒 Viewing as VIEWER — log in as ADMIN to modify settings.

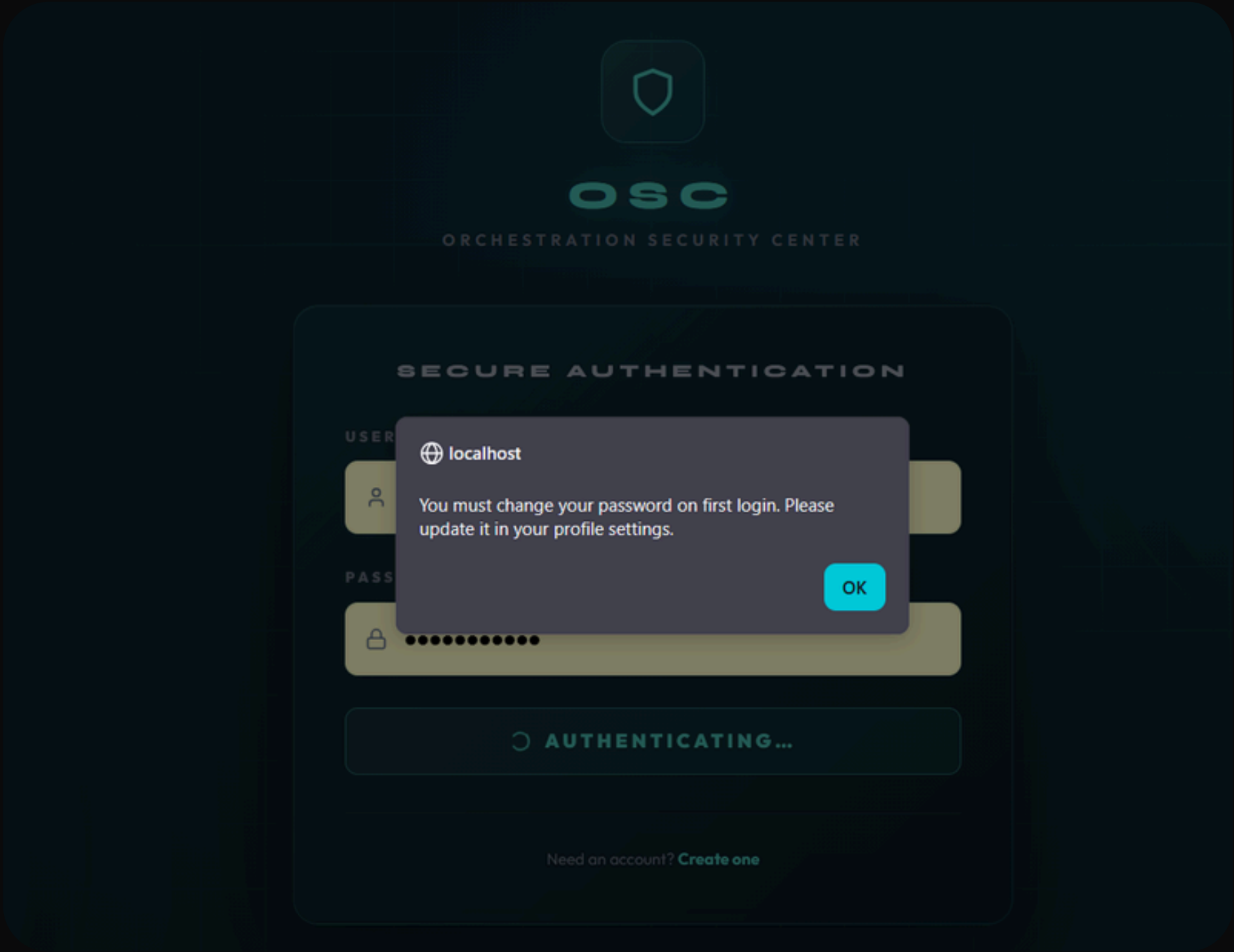
 SYSTEM CONFIGURATION
Read-only view — admin login required to modify settings

 OpenVAS Scanner Network vulnerability scanner (GVM/GMP) ON	 SIEM Integration Elasticsearch + Wazuh event pipeline ON
 LLM AI Validation Gemini triage of findings (never overrides reprobe) ON	 Nmap Scanning Port and service discovery ON
 Nuclei Scanning Template-driven vulnerability scanner ON	

ANALYST PRIVILEGE

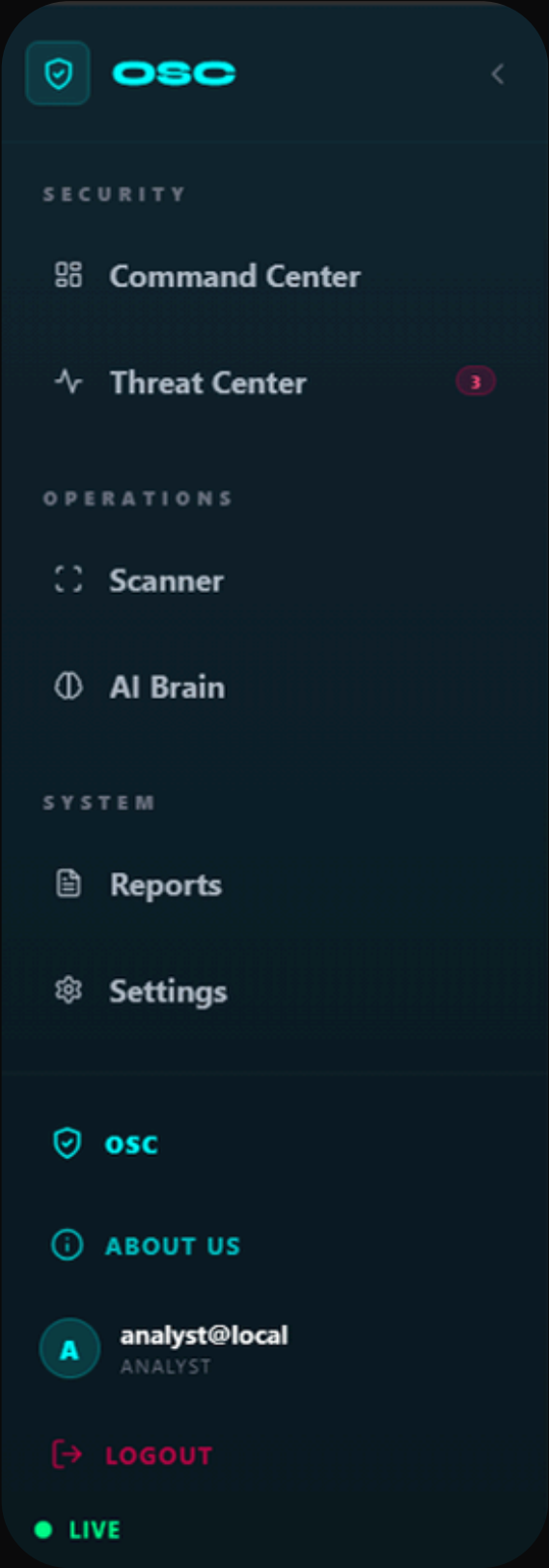
ANALYST PRIVILEGE

PASSWORD CHANGE



ANALYST PRIVILEGE

1.LEFT NAVBAR



ANALYST PRIVILEGE

2.PROFILE

MY PROFILE

A

analyst@local

analyst@local

ANALYST

PNG or JPEG, up to 2 MB.

PERSONAL INFORMATION

EMAIL

analyst@local

FULL NAME

Jane Doe

PHONE

+1 555 555 5555

BIO

Tell your team a bit about yourself...

SAVE CHANGES

CHANGE PASSWORD

CURRENT PASSWORD

NEW PASSWORD

CONFIRM NEW PASSWORD

UPDATE PASSWORD

ANALYST PRIVILEGE

3.ADVANCED SCAN - LAB ENVIROMENT DISABLED

 NEW SCAN

1 TARGET

>

2 PROFILE

>

3 SCHEDULE

>

4 REVIEW

 **Target** — Who are we scanning?

Existing target

Manual URL / IP

 Lab Environment ADMIN

https://example.com or 10.0.0.1 or 192.168.1.0/24

URL, IP, hostname, or CIDR range.

STANDARD · 2 tools

Cancel

Continue

ANALYST PRIVILEGE

4.NODE DELETION DISABLED

Command Center

Operations

Threat Center

AI Brain

Reports

Settings

SCAN

HISTORY

NODES

LAB

Scan Targets

Asset Discovery

Add Target

[Lab] Corporate API Gateway

http://lab_api_gateway:8081

LAB

Unknown stack

Scan...

AI

admin only

[Lab] E-Commerce Web Server

http://lab_webserver:3000

LAB

Unknown stack

Scan...

AI

MGMT 10.10.40.0/24

10.10.40.0/24

LAB

Unknown stack

Last scan: 5/23/2026

Scan...

AI

ANALYST PRIVILEGE

5.VULNERABILITIES ACCEPTED RISK DISABLED

☐

CRITICAL

PostgreSQL COPY PROGRAM Allows OS Command Execution (#04)

postgresql: //lab_database:5432/prod

The PostgreSQL superuser can invoke COPY ... FROM PROGRAM to execute arbitrary shell commands on the database host.

CVSS 9.9

CONFIDENCE 98%

LAB_MEDIAN

In Progress

Open

In Progress

Fixed

False Positive

Accepted (Risk) — admin only

ANALYST PRIVILEGE

6.SETTING

FUTURE WORK

Take Action Automatically

Auto-block a malicious IP or isolate a compromised host the moment a critical threat is confirmed.

Email Reports & Alerts

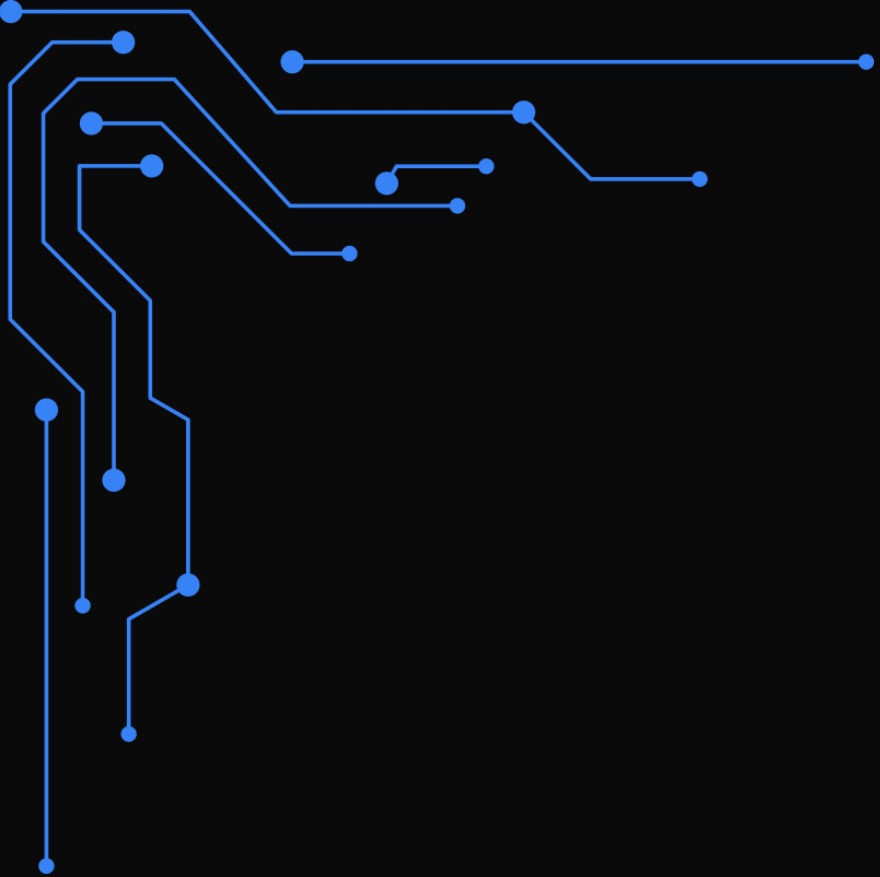
Stakeholders get the signed PDF report and critical alerts by email, no login needed.

Dark Web Monitoring

Automatically check if company emails or passwords have leaked in known data breaches.

AI Chat Assistant

Ask the AI advisor plain questions like “what should i fix first?” instead of just reading a report.



THANK YOU